SYLLABUS and PLAN FOR THE COURSE
MA1101 : Calculus

Course Details

Functions of two and three variables:
Regions in plane, level curves and level surfaces, limit, continuity, partial derivatives, the increment theorem, directional derivatives and gradient, tangent planes and normal lines, Taylor's formula, extreme values, Lagrange multipliers, Volumes by slicing, Volumes of solids of revolution, cylindrical shell method, double integral and iterated integral, double integral in polar coordinates, triple integral, triple integral in cylindrical coordinates, triple integral in spherical coordinates, change of variables.

Series Expansion of Functions:
Series, Taylor series, Fourier series, odd and even functions, half range Fourier series, Functions of arbitrary period.

Course References:

Text:
[T] J.R. Hass, C.E. Heil, M.D. Weir and P. Bogacki, Thomas' Calculus: Early Transcendentals, 15th Edition, Pearson Education, 2023.
[P] N. Piskunov, Differential and Integral Calculus Vol. II, Mir Publishers, Eng. Trans.1974.

References:
1. E. Kreyszig, Advanced Engineering Mathematics, 10th Ed., John Wille & Sons, 2010.
2. S.T. Tan, Calculus, Cengage Learning, 2010.
3. N. Piskunov, Differential and Integral Calculus Vol. 1 & II, Mir Publishers, Eng. Trans.1974.

Details of the plan for the course

Quiz – I : Text Book [T]: Chapter 6 (Sections 6.1-6.2) and Chapter 14 (Sections 14.1 – 14.9) (Quiz–I syllabus with assignment sheet–1 may be completed in 15 L). Quiz I Date - 24 February 2026.

Quiz – II : Text Book [T]: Chapter 15, Sections 15.1 – 15.5 and 15.7-15.8
(Quiz–II syllabus with assignment sheet–2 may be completed in 14 L). Quiz II Date - 24 March 2026.

After Quiz – II : Text Book [P]: Chapter 4 (Sections 4.1-4.2, Sections 4.16-4.17, Sections 4.19-4.20), and Chapter 5 (Sections 5.1-5.6)
(Remaining syllabus with assignment sheet–3 may be completed in 11 L). End-Sem Date – 07-May-2026.